

ECE 313: Probability with Engineering Applications

2025 Fall Instructors: Piao Chen & Xu Chen

Homework 6

Name: _____

Student ID: _____

Due Date: November 14 23:59, 2025

Problem 1. A box contains 5 red and 5 blue marbles. Two marbles are withdrawn randomly. If they are the same color, then you win \$1.10; if they are different colors, then you win $-\$0.90$ (i.e., you lose \$0.90).

- (a) Calculate the expected value of the amount you win.
- (b) Calculate the variance of the amount you win.

Problem 2. Bob plays a lottery game. He purchase a lottery ticket for this game each week. Suppose the probability of winning a \$5 prize on each ticket is $1/20$, and the probability of winning the grand prize on each ticket is $1/200,000,000$.

- (a) Bob will purchase a total of 52 tickets in the year 2025 (=52 weeks). What is the expected number of times that Bob wins a \$5 prize this year?
- (b) Find the expected number of tickets needed until a grand prize is won by Bob (suppose he could play the game forever).

Problem 3. Assume the number of students arriving to the class in an interval of t minutes, N , is Poisson distributed with mean rate of 3 students per minute. Let T represent the inter-arrival times of students.

- (a) Write the probability mass function (pmf) of N .
- (b) What distribution best describes the inter-arrival times of the students? Write the probability density function (pdf) of T .
- (c) Calculate the variance of the number of students arriving in a period of 10 minutes.
- (d) Calculate the variance of the number of minutes until the arrival of the next student.

Problem 4. Let the random variable X be uniform on the open unit interval $(0, 1)$. Let

$$Y = g(X) = -\ln(X).$$

- (a) Find the pdf of Y .
- (b) Calculate the $E[Y]$ using $f_Y(y)$.
- (c) Now, calculate the $E[Y]$ using

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

- Problem 5.** (a) Assume that you have a random number generator that generates a uniformly distributed random variable X over the interval $[-3, 5]$. Find a linear function g such that $g(X) = Y$, where Y is uniformly distributed over $[0, 2]$.
- (b) Solve part (a) with $X \sim N(-2, 4)$ and $Y \sim N(1, 16)$.